Meeting on Farmboy, Poughkeepsie. Nov 5, 1960

Herwitz
Sayer
Oldfield
Kolsky
Fran Allen
Dick May
Jack Bisson

alpha language:

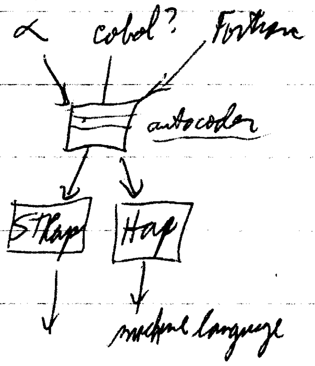
CEIR + computer usage bid .5
IBM 1,9 total bid .9 extra,

alternatives:

Full system 1,9 Jun'62
Intermediate 1,5 Dec'61 all but implicit data def
small 1,3 Sep'61 autocoder + little more

alpha to assembly (no intermediate step)

would like:



But customer is not interested in cobol or Fortran.

Source language ~90% defined.

NSA: Joe Blum — insists on their def. of α. (Schmidt's version dead)

- an updated manual is nearly done -- } Their manual oriented toward problem
- data definition .. all details have not } programmer -- not coder.
been written down.

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Hap: 1 → 1 statements to abs. machine language,
autocoder: 1 → many - function dependent
alpha: an analysis phase

Planning group has been working on data definitions .

"string" - "cord" - "ropes"

string: homogeneous - all elements same length, etc.   eg. vector

cord ;   vector + heading .

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Integers start on 8 bit multiples,

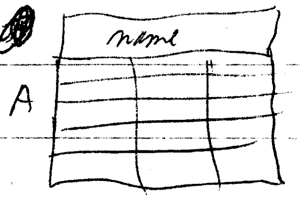
data object - each has name.

- largest starts on full wd boundary

inhomogeneous -- needs literal subscripts,

so one refers

B = name + A   A = B(2, I, J)   B ≡ cord



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Data Statements:

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B   FORM   , NAME, A
A   DEFINE , $F(5,3)
```

store forward, row wise

Alpha would not necessarily have to have these it would

make Form, Define, Do statements from other

Phase III } "Data gleaning" pass -- makes table of Form, define
auto coder → HAP)

Phase III -- all passes not defined yet.

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Example of α:   C = ADD(A, B)   — one need not write out def of C

A \ $FLT(4,3,2)   and indexing of individual steps.

Fran Allen
Ann Ewing
Marcl Orr
John Carter
} planning group

Geo. Grover
C. Gerbrach
P. Herwitz
} review & concur

"Field Testing" begins Sep '61, Dec '61, etc.

How long? several months?

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α - language of "1 shot programs"

GFO - regular runs,

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6 people on autocoder
+3 people on HAP II,

have now done (basic tools)
{ "basic autocoder"
- simple substitution of macros,

- a "shell" can execute SOS type macros.

- should do some subroutine handling

all compiling logic under general subroutines

~ autocoder is essentially alpha?

Question: how far along is autocoder?

which macros will be ready, when,

— Fortran macros will come first (Fortran macros are smaller subset)

- complexity of system? -- can be maintained & changed easier.

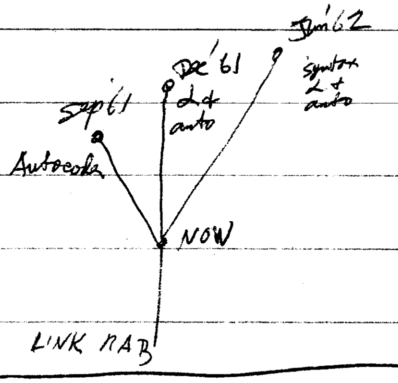
- paying for commercial devel?

- time scale of program?

by pooling job can cut down effort? -- basic assumption,

planning complexities may over do this gain ---

- planning committee has been working on autocoder language -- too much like α source language?



— note decision must be made now
i.e. Dec'61 path will not incl autocode in Sep

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Question of intermediate milestones;

α not to be judged as production effort . . . . reduces chance of backing off.

- interpretive language internal to compiler,
- complete subroutinizing -- is that good?
- schedule?

{ combining α + Fortran — basic decision
{ "Layering" of languages —

~25K HAP instrs
+22
~50K

2000 instrs per man year

(see schedule!)

1. There will be no Host for 3 mos
2. If a move takes place -- lose some people + ~4 to 6 ~~[illegible]~~ per day.

Planning Phase is not on schedule now.

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Tue p.m. - Sayre + I
Fri a.m. - Meeting - Bethesda

Dec 6 Poughkeepsie

(Brannin) during past 2 mos.

- planning has been bottleneck
- strictly substitution macros have been written + debugged. – I/O STRETCH MCP, + HARVEST descriptors

autocoder – language is one for one -- nothing implied
in form: operator, operand$_1$, operand$_2$, with, operand$_3$

Q: – descriptors? are these written
list of statement?

(Parsons) alpha statements are defined

I/O macros

eg alpha "give me a file"
↓
autocode read, write, skip, read ...
↓
~~[illegible]~~ STRAP
↓
machine,

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25 Farmboy Manyears;
25 × $20K ≈ $500K + computer time.

1/3 going into alpha?

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discrepancy on schedule?

autocoder Sept 61 or June 61?
self contained system
does not ~~[illegible]~~ incl auto descriptors to handle α.

(Gerleich) -- of slow start on planning --

Milestones:

-Report on source lang. -- symbols, syntax, terminology,
character representation autocoder level,
data definition -- now finished report.

3 phases of autocoder (time phase)
(1) Subset, macros, only
(2) Data def to tables for further use
(3) "assume" instructions define space, assume space, overlap space
rules for "holes" in data (Indexing phase)

Operating phases:
(1) Syntax
(2)
} → reduce to autocoder, & creation of table
(3) Expanding to machine code. ← (most of effort here now)

a list of autocoder operations -- not now in existence

Descriptors 2 categories: haven't started on indexing yet.

Q: What about going straight to STRAP?
- expandability would be lost.
- manpower on STRETCH would not be available (10 people) on autocoder.

Autocoder will go on regardless as to whether alpha goes on

(Gerbind) - Alpha doesn't do well what machine does well -- it has to be openended.

What will be presented to customer

(cont'g.) no fat in orig. proposal – schedule or cost..

if it is too late we could propose sooner

These would make big diff { data simplification
outline language (data segmentation)

– can be done by Jan '62   ie, must [illegible] do own segmenting
& defining of data (Fortran-like)

would not prejudice full [illegible] program later.

{ if syntax phase I is removed..
could be done by ~Sept '61

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Plan: Present #1 and #2 (not #3 unless there is serious prob),
no competitor can really better offer.

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(Dunwell) re Thms.

There was meeting set up for Dec 8 sometime ago with McWhirter, Tordella, et al

—a prelim meeting was held last week at Ft Meade

- acceptance tests

schedules, -- will not pull out machines

What will be said about Farmboy? 30 min. a decision has got to be made by Jan 1 - pt of decision (a letter has been sent already) to deviate from full α,

apr '61 - money runs out.

3 alternatives min - appl. autocoder - Oct 1 + 90 days field testing to useful work

inter - lim. alpha - Jan 1 + 90 days

other nos. have not been mentioned } → ($1,925) full - full alpha - apr 1 + 90 days

HAP I, HASTI, HCP, 7049 Sim., , STR Progs.

they are doing

They will try for full alpha. (CEIR will give more like compromise α.)

—not a proper meeting to raise decision choices

No alpha - (some other contractor) $1,000 (incl 6-7 people to end of 62)

Compromise α ~1.7

Full α - (a step toward Lightening) $1,925 ← about $600K is cost of autocoder

⇒ a lower price should be given? no chance by Thms. -- must get to Harmon & Dolando,

with 12 people only -- it is better to use auto coder people, — Harmon There Thaws?

green, McCall, Soderburg

Tordella, Barlow, McLaren, Gallman, Bachelor, Molton

SGC Dec 4

Dave Sayre - Mon 9am

Karl

HGK

PHenitz

Burnett - MIT non-pr.

String Transformation language.

~~6 Ave. 51st~~

Time + Life Bldg 9:45

4th floor classroom

Hotels: Wellington - AWRE

~~Seymore~~

Manhattan - Seymore

Taft - LASL

Bruce Oldfield Dec 6

Mike Kelly - problem with customer - Time critical - will make decision on negative to us now

Al Harmon - will review Farmboy in NY? who?

Ross Green - will talk to Harmon -

— all to [illegible] Fri

decide on proposal then

→ What is going to be proposal to customer? What alternative.

(1) What to get NSA approval.

(2) How do we implement?

SGC Dec 8 pm

meeting today

NSA = "Scorched" whole program

Harmon - spoke on Farmboy

- we would make presentation week after next

Harm - Jacobs, phone call

next Thur

H.C.G.I at Pok

& get J's advice

Fortran directly is easier

IBM pick up tab on autocoder

NSA doesn't realize cost of total system

→ B. Oldfield wants me to get copy of α-papers + keep up to prepare for another review

Thrush Mgmt - OK

- approach

Team - capable

sched - ?

Recommend at present time:

(1) more milestones - closer control on progress

(2) Consider IBM paying for autocoder

(3) another review in ~ end of Feb '61

2nd. [illegible]